Silverline House — 25-Storey Residential HRB

Transfer Slab Design Note

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1 Introduction

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings. A post-tensioned transfer slab at Level 1 spans between the core walls and the perimeter columns at 7.5 m centres, supporting eight residential floors above. The slab is 600 mm deep with an unbonded tendons in both directions at 200 mm centres. The design has been analysed using finite element software with validation by hand calculation for critical sections. Shear checks at column heads comply with BS EN 1992-1-1 clause 6.4 with punching shear reinforcement provided at all columns. The transfer slab is classified as a Critical Structural Element for inspection purposes; the inspection and maintenance strategy is defined in the Fire and Emergency File.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Design Actions

Design loads have been derived in accordance with BS EN 1991-1-1 and BS EN 1991-1-3 for imposed, snow and wind loading respectively, with the UK National Annex applied throughout. The imposed load for residential floors is 1.5 kN/m² with an additional 1.0 kN/m² for partitions. Plant rooms at roof level are designed for 5.0 kN/m². Wind loading has been calculated for a basic wind speed of 23 m/s (10-minute mean at 10 m height in open country) adjusted for the site location and topography in accordance with BS EN 1991-1-4 and the UK National Annex. The net lateral wind force on the building is 3,200 kN at the 50-year return period, resisted by the reinforced concrete core. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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3 Analysis and Results

Lateral stability is provided by a reinforced concrete core of plan dimensions 12 m × 8 m positioned at the centre of the building footprint. The core walls are 300 mm thick below Level 5, reducing to 250 mm above Level 5 and 200 mm above Level 15. Core wall reinforcement has been designed for combined bending, shear and axial forces using the General Method of BS EN 1992-1-1. The inter-storey drift under the 50-year wind load is limited to H/500 = 17 mm to satisfy serviceability requirements. The building has been checked for progressive collapse in accordance with BS EN 1991-1-7 Annex A, adopting the notional member removal procedure for all floors above ground level. All reinforced concrete elements are specified in C32/40 concrete (characteristic cylinder / cube strength) with a maximum water-cement ratio of 0.55 and minimum cement content of 300 kg/m^3. Post-tensioned slab tendons are specified in low- relaxation 15.7 mm strand to BS 5896, stressed to 75% of characteristic tensile strength. Reinforcement throughout is Grade B500B to BS 4449. Concrete cover to reinforcement is 35 mm to internal elements and 50 mm to external elements exposed to the weather, in accordance with BS EN 1992-1-1 Table 4.4N for exposure class XC3. Minimum cement content and maximum water-cement ratio comply with the durability requirements of BS 8500-1 for the intended service life of 50 years.

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